

$$\Rightarrow \chi^2 = \sum_{i=1}^N \left(\frac{y_i - y(x_i, a, b)}{\sigma_i} \right)^2$$

The optimization condition implies

$$\frac{\partial \chi}{\partial a} = \frac{\partial \chi}{\partial b} = 0$$

Using $y(x_i, a, b) = a + x_i b$

$$\Rightarrow \frac{\partial \chi}{\partial a} = \frac{1}{2\chi} \frac{\partial \chi^2}{\partial a} = 0$$

$$\Rightarrow \frac{\partial \chi^2}{\partial a} = 0$$

$$\frac{\partial \chi^2}{\partial a} = \sum_{i=1}^N 2 \left(\frac{y_i - a - x_i b}{\sigma_i^2} \right) = 0$$

$$\frac{\partial \chi^2}{\partial b} = \sum_{i=1}^N 2 \left(\frac{y_i - a - x_i b}{\sigma_i^2} \right) x_i = 0$$

We can factor out 2 so we get conditions

$$\sum_{i=1}^N \left(\frac{y_i - a - x_i b}{\sigma_i^2} \right) = 0$$

$$\sum_{i=1}^N \left(\frac{y_i - a - x_i b}{\sigma_i^2} \right) x_i = 0$$

We may expand the sums as

$$\sum \frac{y_i}{\sigma_i^2} + \sum \frac{-a}{\sigma_i^2} + \sum \frac{-bx_i}{\sigma_i^2} = 0$$

As a & b are constants

$$\Rightarrow -a \sum \frac{1}{\sigma_i^2} - b \sum \frac{x_i}{\sigma_i^2} + \sum \frac{y_i}{\sigma_i^2} = 0$$

From the second equation

$$-a \sum \frac{x_i}{\sigma_i^2} - b \sum \frac{x_i^2}{\sigma_i^2} + \sum \frac{y_i x_i}{\sigma_i^2} = 0$$

As x_i, y_i & σ_i are known values we can calculate a & b

denote $\sum \frac{1}{\sigma_i^2} = \sigma$; $\sum \frac{x_i}{\sigma_i^2} = x$; $\sum \frac{y_i}{\sigma_i^2} = y$; $\sum \frac{x_i^2}{\sigma_i^2} = x^2$; $\sum \frac{x_i y_i}{\sigma_i^2} = xy$

$$\Rightarrow \left. \begin{aligned} y &= a\sigma + bx \\ yx &= ax + bx^2 \end{aligned} \right\} \begin{aligned} a &= \frac{y - bx}{\sigma} \\ a &= \frac{yx - bx^2}{x} \end{aligned}$$

$$\Rightarrow \frac{y - bx}{\sigma} = \frac{yx - bx^2}{x}$$

$$\Rightarrow \cancel{yx} (y - bx) - (bx)x = \sigma yx - \sigma bx^2$$

$$\Rightarrow (y)(x) - \sigma yx = (bx)x - \sigma bx^2 = b((x)(x) - \sigma x^2)$$

$$\Rightarrow b = \frac{(y)(x) - \sigma yx}{(x)(x) - \sigma x^2} //$$

$$\Rightarrow a = \frac{y - bx}{\sigma}$$

$$\Rightarrow a = \frac{1}{\sigma} \left(y - (x) \frac{(y)(x) - \sigma yx}{(x)(x) - \sigma x^2} \right) //$$

HW8 - Computer Simulation II

Schewtschik, Guilherme (3748052)

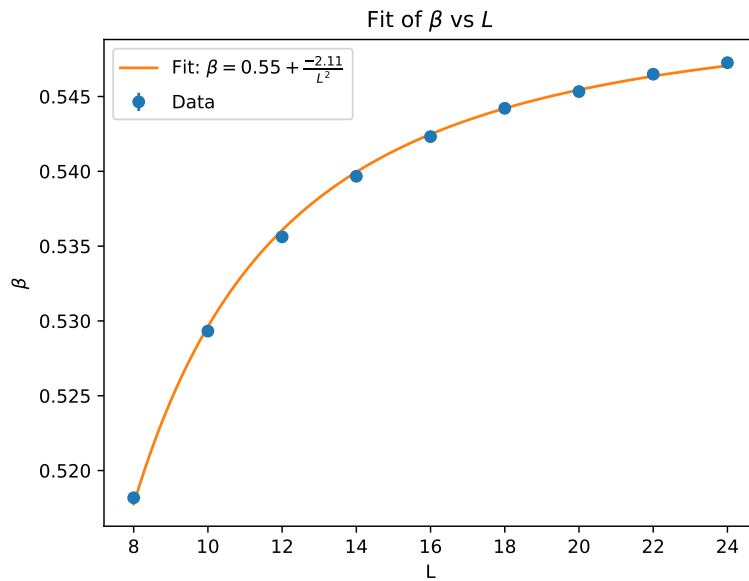
June 9, 2026

1

Given that beta and L are related by the equation:

$$\beta = a + \frac{b}{L^2} \quad (1)$$

It is best to work with the variable $\tilde{L} = 1/L^2$, and then convert it back to L . Fitting a and b for $L_{min} = 8$ we get the following graph:



We can see that the fit is at a first glance good but deviates in some points, lets calculate the relative error of the individual points and $\chi^2/(N - 2)$:

$$\frac{\chi^2}{N - 2} \approx 10.44$$

β	β_{fit}	Relative Error
0.51818	0.51776	4.22
0.52932	0.52962	-3.03
0.53562	0.53607	-4.07
0.53967	0.53995	-2.84
0.54232	0.54248	-1.95
0.54421	0.54421	0.05
0.54533	0.54544	-1.41
0.54649	0.54636	2.64
0.54726	0.54705	2.94

Table 1: Fit results for all lattice sizes $L = 8, \dots, 24$.

Given that the ansatz expects $\chi^2/(N - 2) \approx 1$, the fit quality is poor for $L_{min} = 8$, now ranging over L_{min} values we can find the optimal range of L values to fit the data:

$L_{\min}$	N	a	b	χ^2/dof
8	9	0.550716	-2.109338	10.443
10	8	0.550936	-2.181172	2.859
12	7	0.551079	-2.234493	1.143
14	6	0.551122	-2.251716	1.268
16	5	0.551162	-2.269470	1.584
18	4	0.551242	-2.307417	2.210

Table 2: Weighted fits for different lower cutoffs $L_{\min}$.

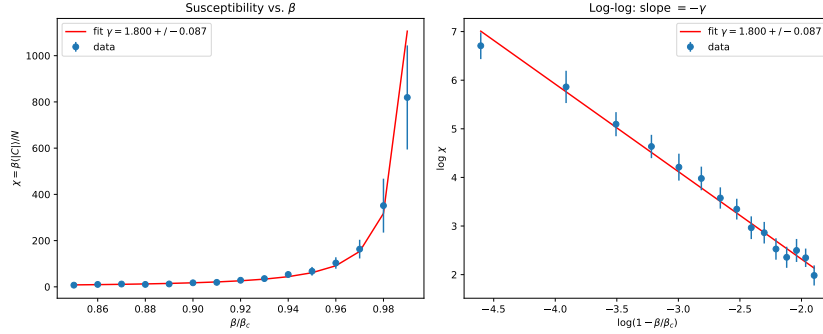
So for $L_{\min} = 12$ we get the best fit, with $\chi^2/(N - 2) \approx 1.143$, and the fit parameters are $a \approx 0.55$ and $b \approx -2.23$. This means that the critical beta value in the thermodynamic limit is approximately $\beta_c \approx 0.551079$.

2

Rewrite the equation in a logarithmic form:

$$\ln(\chi(\beta)) = -\gamma \ln(a) - \gamma \ln(1 - \beta/\beta_c)$$

Fitting the curve in log space for $\beta \in [0.99, 0.85]\beta_c$ and taking into account the error $e_{\ln(\chi)} = \sigma/\chi$, where sigma is the standard deviation from cluster size, we get the following plots.



The values for $\gamma = 1.80 \pm 0.087$ is well in the theoretical expectation of $\gamma = 1.75$ for the 2D Ising model.